

* Internal / ** External

Design Specification**Communication specifications**

* Machine * Power ** Software * System

Product
name

ADAMS A1c HA-8180V

Managed by

Product Development
DivisionLaboratory Product
2 Team

Preparation date	Approval	Check	Author
Jul 28, 2009	Takagi	Yoshioka	Yoshida

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A	Jan 18, 2010	Takagi	Egawa	Yoshida
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Revision record

Revision No.	Revised date	Item	Description / timing of implementation
C	Jun 22, 2011	2.6.4 Measurement result message	Clerical error of item names "^^^PeakS" -> "^^^PeakHbS" "^^^PeakC" -> "^^^PeakHbC"
D	Aug 28, 2013	2.4.5. Notes for designing the PC application	Addition of notes for designing the PC application.
		2.6.4.Measurement result message [Bit array for error information] (※3)	Correction of errors. Deletion of "Press move"

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1. Overview

This document describes the HA-8180V communication specifications (RS-232C) on the HA-8180V format compliant with ASTM. Information on HA compatible formats (HA-8121 to HA-8160) is not included. Refer to the relevant specifications for details on the compatible formats.

2. Interface specifications

HA-8180 adopts the communication protocol compliant with ASTM-1381 (ARKRAY standard). Note that the HA-8180 standard format differs from the compatible formats in the respects including the transmission method, number of re-transmission and time-out.

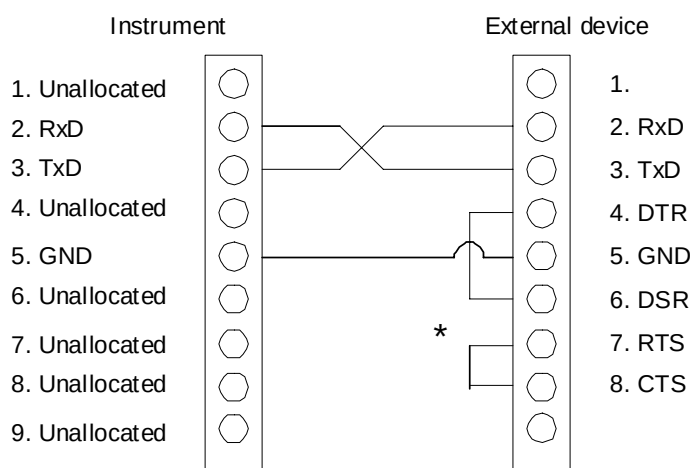
2.1. RS-232C communication settings

Item	Description
Interface	Voltage type (RS-232C compliant, bit serial output)
Transmission method	Asynchronous communication method Half-duplex simplex alternate communication
Transmission rate	Selectable from 300, 600, 1200, 2400, 4800, <u>9600</u> , 19200 bps
Start bit	1 bit
Data bit	8 bit
Parity bit	None
Stop bit	1 bit

Note 1) “Start bit” and “Stop bit” represent the binary digits “0” and “1”, respectively.

Note 2) The default value is underlined.

2.2. Schematic diagram (Instrument / External device) – Connector pin Nos., signals, cables



* When the CTS and RTS terminals are used on the external device, connect them in the cable.

* When the DSR and DTR terminals are used on the external device, connect them in the cable.

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2.3. RS-232C communication frame structure

2.3.1. Frame structure

STX	FN	Transmission data (Max. 240 bytes)	ETB / ETX	CSH	CSL	CR	LF
-----	----	---------------------------------------	--------------	-----	-----	----	----

2.3.2. Maximum frame length

The maximum frame length is 247 bytes, including STX, ETB or ETX, Checksum, CR and LF. The maximum transmission data length is 240 bytes (247 - 7). If the transmission data length exceeds 240 bytes, it will be sent in several frames.

2.3.3. Frame control code

Code	Hexadecimal	Description
STX	02H	Start of text (frame)
FN	30H - 37H '0' - '7'	Frame No. Begins with "1" and incremented by one. Returns to "0" after reaching "7".
Transmission data		Communication text
ETB	17H	End of transmission block
ETX	03H	End of text (frame)
CSH	Note 1)	Checksum high
CSL		Checksum low
CR	0DH	Carriage return
LF	0AH	Line feed

Note 1) Calculation of checksum

Checksum is calculated by adding up the bytes from after STX through ETX or ETB and dividing the sum by 256 to express its remainder in 2-digit hexadecimal form. The first and second characters of the checksum represent "CSH" and "CSL", respectively.

<Example>

The sum of the bytes after STX to ETX = 1000

$1000 \bmod 256 = 232 = \text{E8H}$

CSH = "E" (45H) , CSL = "8" (38H)

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2.4. RS-232C communication protocol

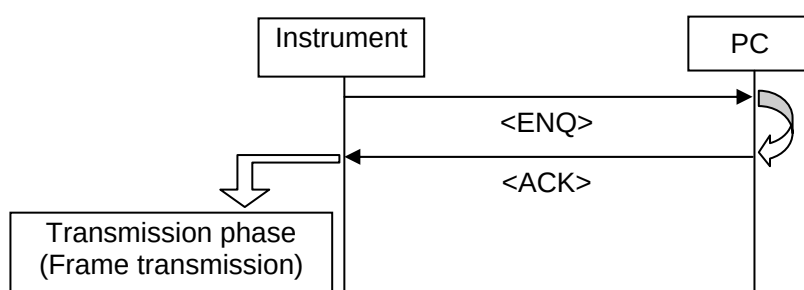
2.4.1. Communication control code

Code	Hexadecimal	Description
ENQ	05H	Enquiry (Request for start)
ACK	06H	Acknowledgement
NAK	15H	Negative acknowledgement
EOT	04H	End of transmission and request for receive interrupt

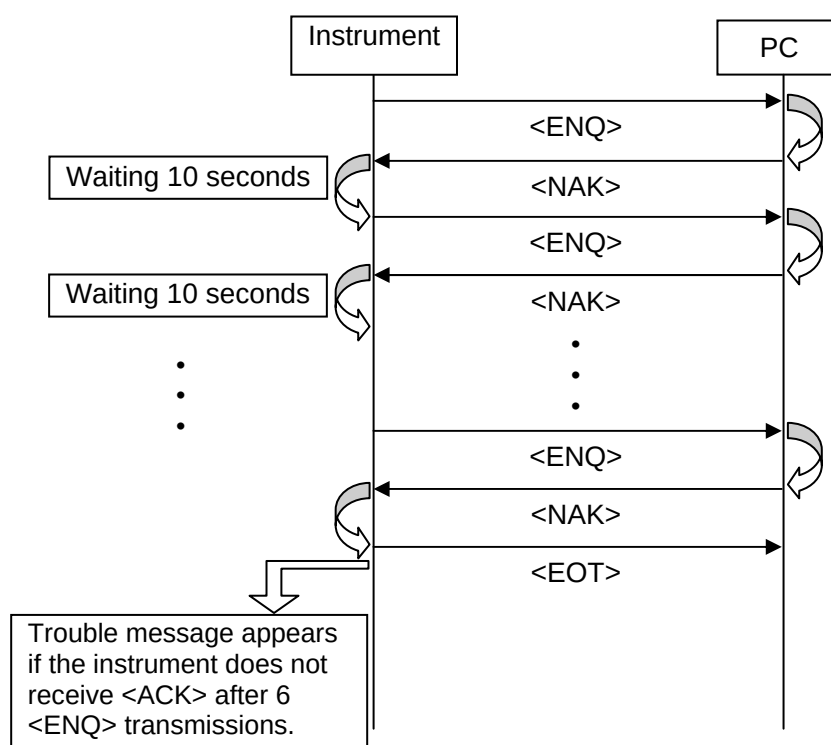
2.4.2. Establishment phase

(1) Instrument <ENQ> transmission

- The instrument transmits <ENQ> when it has a message to send.
- If the instrument receives a response other than <ACK> and <ENQ>, it performs the same process as when it receives <NAK>.
- If the instrument receives <ACK>, it moves on to the transmission phase.



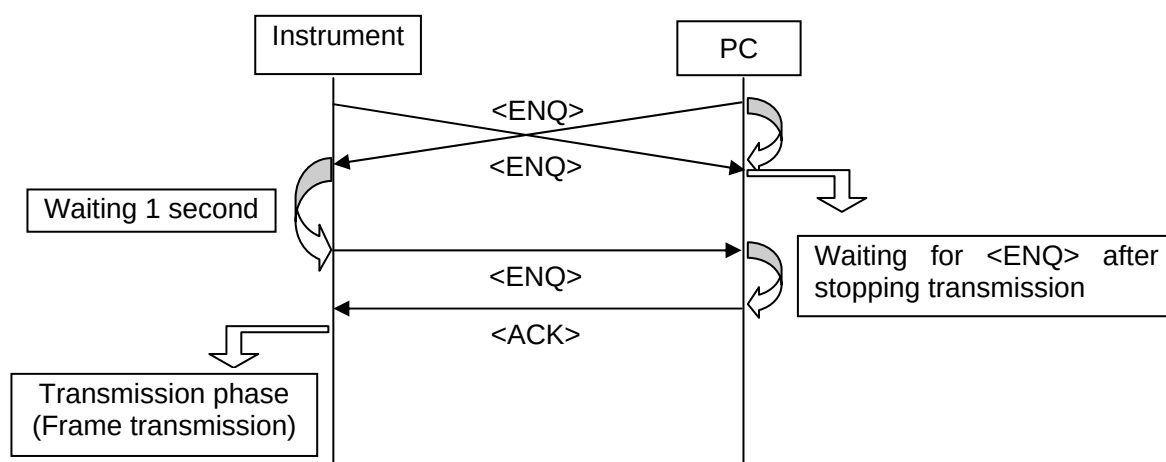
- If the instrument receives <NAK>, it retransmits <ENQ> after waiting for 10 seconds.
- If the instrument does not receive <ACK> after transmitting <ENQ> 6 times (including the first transmission), it sends <EOT> before giving a trouble message. (*)



(*) This is designed to avoid entering into an infinite loop.

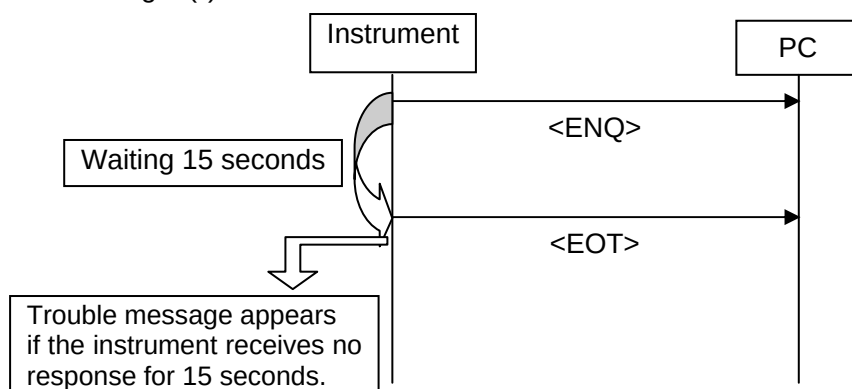
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- If the instrument receives <ENQ>, it retransmits <ENQ> after waiting for 1 second. (*)
- If the PC receives <ENQ> after sending <ENQ>, it stops transmitting and waits for <ENQ>. If the PC does not receive <ENQ> after waiting for 20 seconds, it retransmits <ENQ>.



(Note) If a collision of <ENQ> occurs, <ENQ> from the instrument is given priority over that from the PC and the PC will wait for <ENQ>.

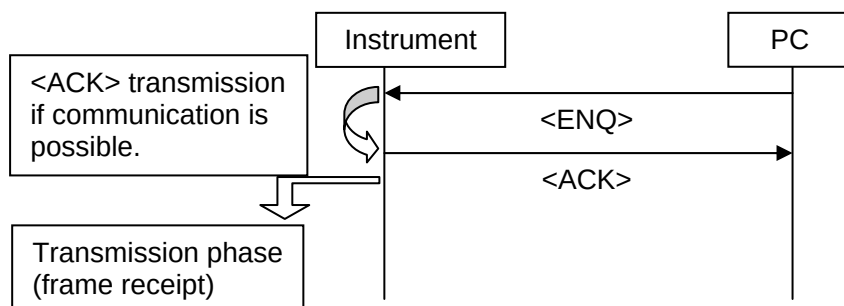
- If the instrument receives no response for 15 seconds, it sends <EOT> before giving a trouble message. (*)



(*) This is designed to detect cable disconnection, including unplugged cables.

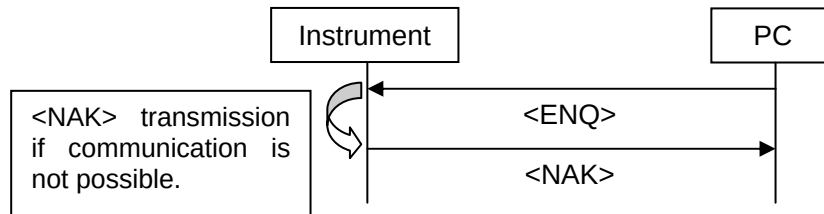
(2) Instrument <ENQ> receipt

- After receiving <ENQ>, the instrument sends <ACK> if communication is possible, and moves on to the transmission phase (frame receipt).



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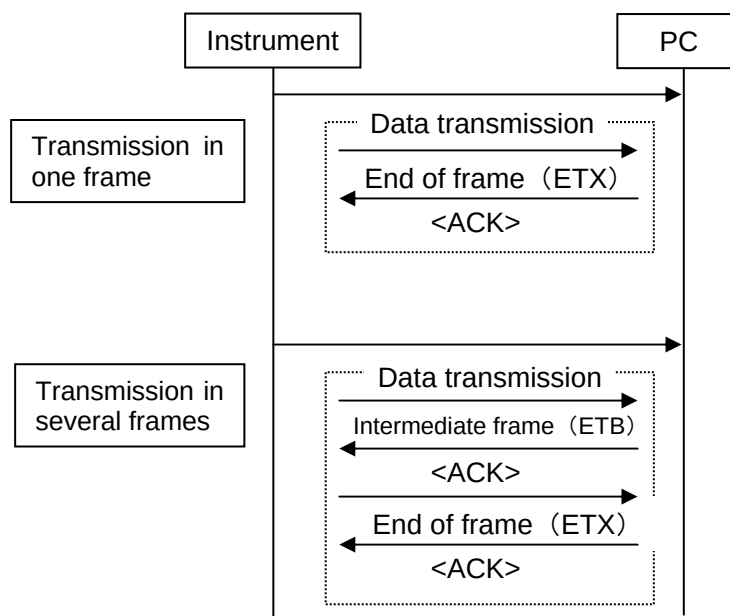
- After receiving <ENQ>, the instrument sends <NAK> if communication is not possible.



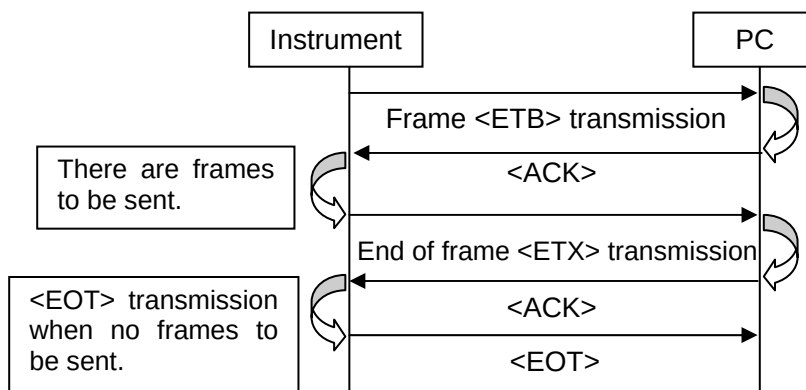
2.4.3. Transmission phase

(1) Instrument frame transmission

- The instrument divides the data into several frames, if it exceeds the maximum frame length.

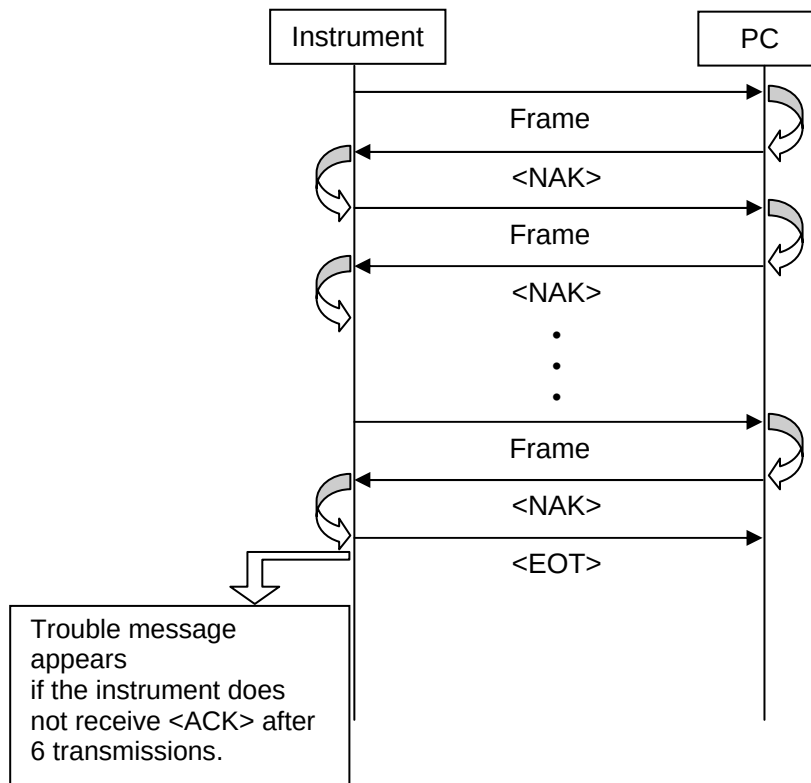


- If the instrument receives a response other than <ACK> and <EOT>, it performs the same process as when it receives <NAK>.
- If the instrument receives <ACK> or <EOT> (*) and there is no frame to send, it transmits <EOT>. If there are any frames to send, it transmits the next frame.
(*) When the instrument receives <EOT> after frame transmission, it regards <EOT> as “Acknowledgement” (<ACK>), not “End of transmission” (<EOT>).
- The instrument transmits <EOT> after completion of all the frame transmissions.



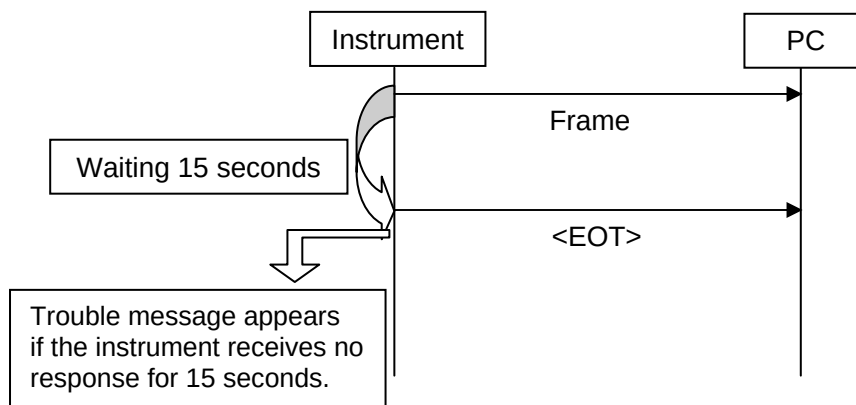
- The instrument retransmits the same frame if it receives <NAK> after sending a frame.
- If the instrument does not receive <ACK> after transmitting the same frame 6 times (including the first transmission), it sends <EOT> before giving a trouble message. (*)

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(*) This is designed to avoid entering into an infinite loop.

- If the instrument receives no response for 15 seconds, it sends <EOT> before giving a trouble message. (*)



(*) This is designed to detect cable disconnection, including unplugged cables.

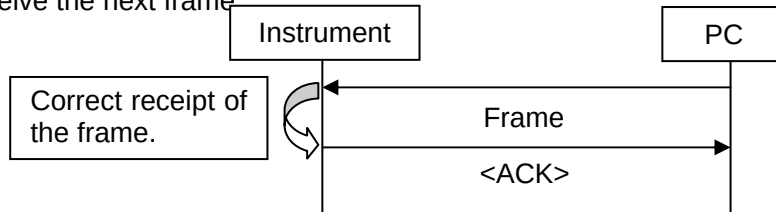
(2) Process before retransmission

If the instrument receives a response other than <ACK> after transmitting data, it will wait for a certain period of time before retransmitting the data. During this period, the instrument disregards any data sent from the PC.

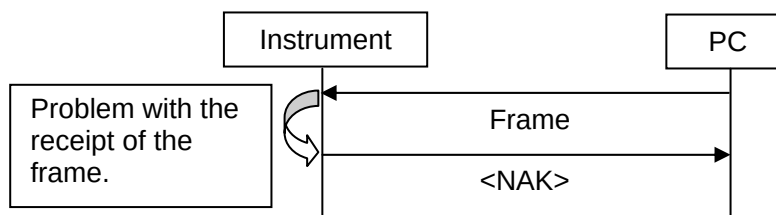
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(3) Instrument frame receipt

- After receiving a frame, the instrument checks whether it has received the frame correctly. If the instrument confirms the correct receipt of the frame, it transmits <ACK> after becoming ready to receive the next frame.



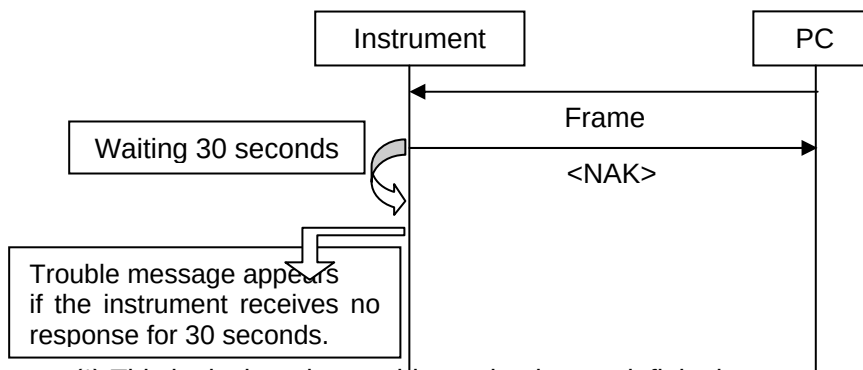
- After receiving a frame, the instrument checks whether it has received the frame correctly. If the instrument finds any problem with the receipt of the frame, it transmits <NAK> after becoming ready to receive the frame again.



(*)

Despite the correct receipt of the frame, <NAK> may be transmitted. If this causes a problem for the PC, lower the setting of the baud rate.

- If the instrument does not receive <EOT> or the receipt completion of the next frame (<CR> <LF> at the end of the frame) within 30 seconds after sending a response, it gives a trouble message. (*)

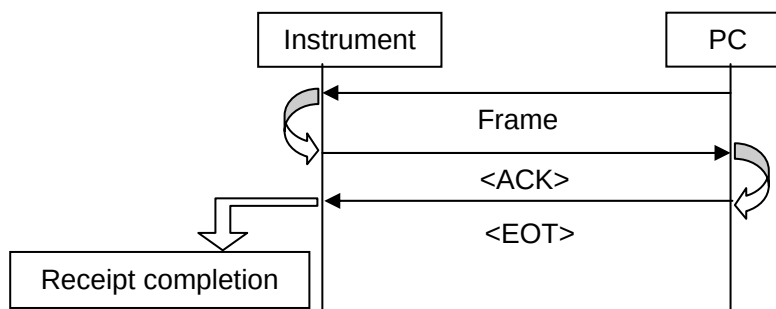


(*) This is designed to avoid entering into an infinite loop.

Criteria for the correct receipt of the frame

- No character errors shall be detected. (Parity error, framing error, etc.)
- Frame checksum shall match the checksum calculated based on the frame received.
- The frame number shall be equal to or one greater than that of the previously received frame.

The instrument determines the receipt completion of all the frames when it receives <EOT>.



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2.4.4. Time-out / Number of retransmission / Waiting time settings

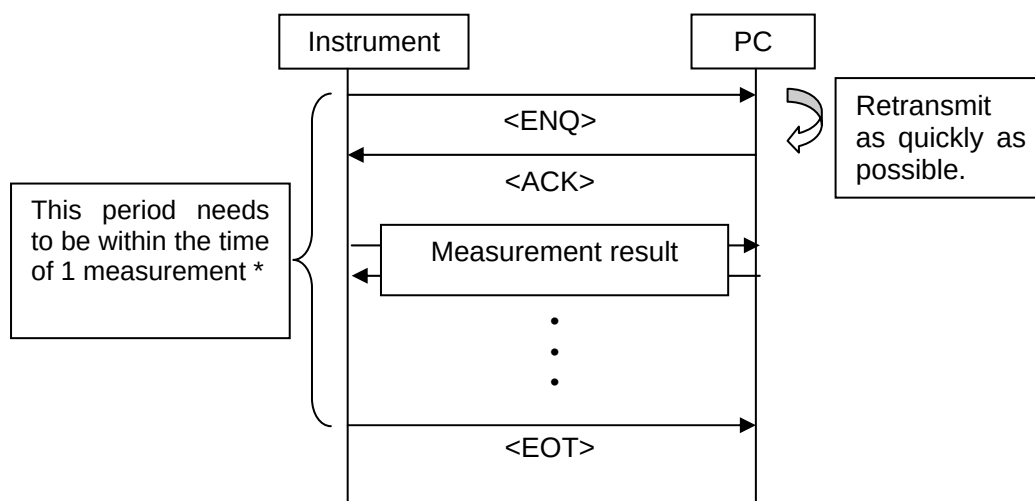
It is advisable to set the following parameters individually. If this is difficult, however, the same value can be used (e.g. setting the same value for the non-response time-out after <ENQ> transmission and frame transmission.)

Parameter	Description	Default
Time-out	Non-response time-out after <ENQ> transmission	15 seconds
	Non-response time-out after frame transmission	15 seconds
	The maximum time between sending a response and receiving <EOT> or the receipt completion of the next frame.	30 seconds
Number of retransmission	The number of retransmission when receiving <NAK> after <ENQ> transmission.	6 times
	The number of retransmission when receiving <NAK> after frame transmission.	6 times
Waiting time	The maximum time waited when receiving <NAK> after <ENQ> transmission.	10 seconds
	The maximum time waited when receiving <ENQ> after <ENQ> transmission.	1 second

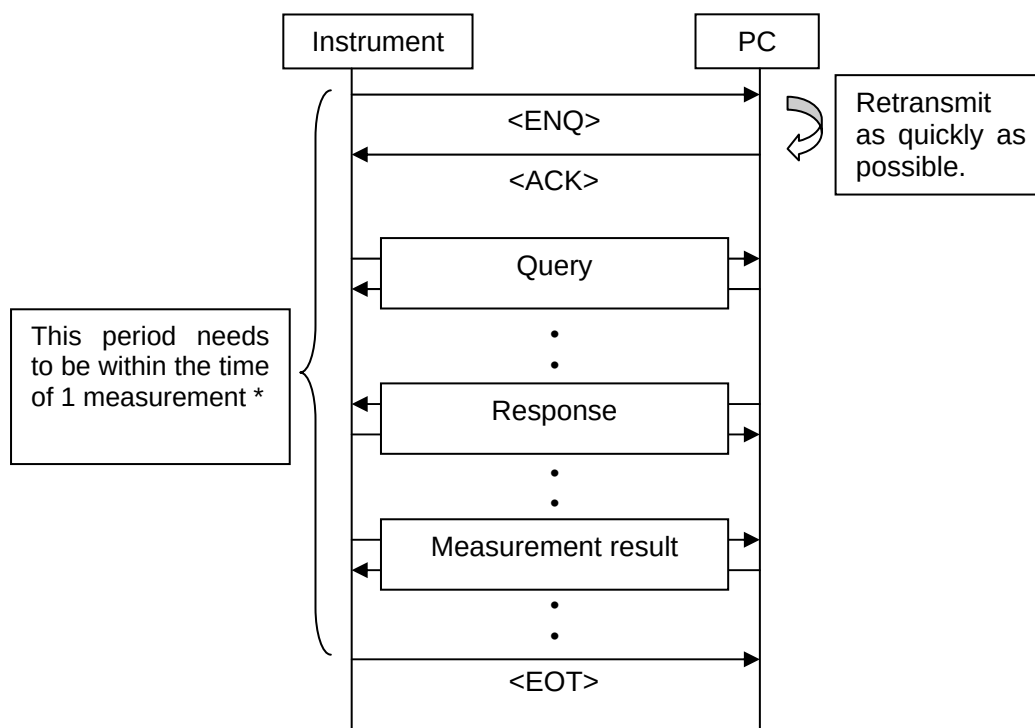
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2.4.5. Notes for designing the PC application

- After receiving data from instrument, PC has to retransmit as quickly as possible.
If responses from PC are late and communication time for 1 measurement result exceeds the measurement time, the instrument keeps sending after the instrument finishes measurement and enter to standby mode.



- If a query function is used, a sequence of communications; query, response and sending measurement result data; has to finish within the time of measurement.
If it exceeds measurement time, the instrument has to insert a skip measurement because it cannot proceed to next measurement and it leads to loss of reagent and time.



*Variant mode:90seconds / Fast mode:48seconds

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2.5. Communication data structure

2.5.1. Message

Data sent and received in communication is called “message”. One message is composed of multiple records in the following hierarchy. The message starts with the header record and finishes with the end record.

Example: Measurement result message

Message header record
Patient information record
Test order record
Result record 1
Result record 2
Message end record

2.5.2. Record

Each record is composed of multiple fields and its content is defined by its record type. The number of fields in the record is fixed according to the record type. The data length in the field is variable (Min. 0 when omitted). Each field is divided by field delimiters. Delimiters are left for the field omitted so as to indicate the field has been omitted. (Delimiters at the end of the record may be omitted.)

Example: Result record

	Field	Example
1	Record type	R
2	Record sequence No.	1
3	Measurement item	HbA1c
4	Measurement value	5.5
5	Unit	%
6	X (Omitted)	No output (NULL)
7	Abnormal flag	ERR.
8	X (Omitted)	No output (NULL)
9	Test result status	F
10	X (Omitted)	No output (NULL)
11	X (Omitted)	No output (NULL)
12	Measurement start date and time	200203041010
	Record terminator	<CR>

R| 1|^^^ValueHbA1c|5.5|%||ERR.||F|||200203041010<CR>

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2.6. Communication format

HA-8180 communication format

- Query message (Instrument -> PC)
- Response message (PC -> Instrument)
- Measurement result message (Instrument -> PC)

2.6.1. Common record

All the above messages include common records (header / end records). This section provides information about the header and end records.

Descriptions of the fields that are defined by "ASTM 1394" and not used in this format have been omitted. Missing field numbers are unallocated fields.

(1) Header record

Header record is composed of 14 fields.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "H"
2	Delimiter setting	1) Set characters Field delimiter Repeat delimiter Component delimiter Escape delimiter	1) " \$^&"
5	Sender name	1) Instrument name: Set characters 2) Instrument No. (S/N): 8-digit number 3) Instrument version: Characters / digits 4) Surplus: Space	1) "HA-8180V" 2) "10900001" 3) "V01.00" 4) " "
14	Communication date / time	1) Communication date / time (YYYYMMDDHHMM): 12-digit number (Year, Month, Date, Hour, Minute)	1) "200607191010"
	End of record	1) End of record (set characters)	1) <CR> (0x0D)

<Example>

H|\$^&|||HA-8180V^10900001^V01.00 ||||| 200203111010<CR>

(2) End record

End record is composed of 3 fields.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "L"
2	Sequence No.	1) Set digit	1) "1"
3	End code	1) Indicates whether measurement has been normally completed. (Normal end : Set character)	1) "N"
	End of record	1) End of record	1) <CR> (0x0D)

<Example> 12-3-

L|1|N<CR>

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2.6.2. Query message

This section describes the query message structure and each record.

▪ Query message structure (Instrument -> PC)

Message header record	H
Query record	Q
Message end record	L

(1) Query record

Query record is composed of 13 fields.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "Q"
2	Sequence No.	1) Set digit	1) "1"
3	Patient ID	1) Barcode ID: 18-digit number/character/symbol ^(*) 2) Port No.: 4-digit number 3) Rack ID: 2-digit number	1) "012345678901234567" 2) "0001" 3) "01"
13	Query code	1) Indicates the content of query (Request an order): Set character	1) "O" (character)
	End of record	1) End of record	1) <CR> (0x0D)

<Example> 12-3-----4567890123-

Q|1|0123456789012345--^0001^01| || || || || || O<CR>

(*) The maximum digit number for the barcode ID is 18. If it is not met, "-" will be used for the lower position.

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2.6.3. Response message

▪ Response message structure (PC -> Instrument)

Message header record	H
Patient information record	P
Test order record	O
Message end record	L

(1) Patient information record

Patient information record is composed of 2 fields.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "P"
2	Sequence No.	1) Sequence number starting with "1"	1) "1"
	End of record	1) End of record	1) <CR> (0x0D)

<Example> 12-

P|1<CR>

(2) Test order record

Test order record is composed of 16 fields, which are different from those of the measurement result message.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "O"
2	Sequence No.	1) Set digit	1) "1"
3	Patient ID	1) Barcode ID: 18-digit number/character/symbol 2) Port No.: 4-digit number 3) Rack ID: 2-digit number	1) "012345678901234567" 2) "0001" 3) "01"
5	Measurement item	1) Measurement item name (either of below) 1) -1. Measure: Set character ^(*) 1) -2. No measure: No output	1) -1. "^^^HbA1c" 1) -2. NULL (No output)
6	Priority	1) Measurement type (either of below) 1) -1. Sample: Set character 1) -2. Control: Set character	1) -1. "R" 1) -2. "C"
16	Sample type	Sample type (either of below) 1) -1. Whole blood: No output 1) -2. Hemolysis: Set character 1) -3. Anemia: Set character	1) -1. NULL (No output) 1) -2. "H" 1) -3. "A"
	End of record	1) End of record	1) <CR> (0x0D)

<Example> 12-3-----45-----6-7890123456-

O|1|0123456789012345--^0001^01||^^HbA1c|R| || || || || ||H<CR>

(*) Measurement item consists of four components but the first three are not used.

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2.6.4. Measurement result message

This section describes the measurement result message structure and each record.

▪ Measurement result message structure (Instrument -> PC)

Message header record	H
Patient information record	P
Test order record	O
Result record	R
Message end record	L

1) Patient information record

Patient information record is composed of the same fields as those of the response message.

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2) Test order record

Test order record is composed of 20 fields, which are different from those of the response message.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "O"
2	Sequence No.	1) Set digit	1) "1"
3	Patient ID	1) Barcode ID: 18-digit number/character/symbol (*1) 2) Port No.: 4-digit number 3) Rack ID: 2-digit number	1) "012345678901234567" 2) "0001" 3) "01"
4	Measurement No.	1) Measurement number: 4-digit number	1) "0001"
5	Measurement item	1) Measurement item name (*2) HbA1c: Set characters	1) "^^^HbA1c"
6	Priority	1) Measurement type (one of below) 1) -1. Normal sample: Set character 1) -2. STAT sample: Set character 1) -3. Control: Set character	1) -1. "R" 1) -2. "S" 1) -3. "C"
16	Sample type	1) Sample type (either of below) 1) -1. Whole blood: No output 1) -2. Hemolysis: Set character 1) -3. Anemia: Set character	1) -1. NULL (No output) 1) -2. "H" 1) -3. "A"
20	Additional comment	1) HbA1c measurement error information 8-digit number (Hexadecimal) 2) Reserve 8-digit number (Hexadecimal) 3) Column degradation information 3) -1. Normal 3) -2. Column degradation 4) Measurement mode 4) -1. Variant mode 4) -2. Fast mode 5) Eluent A: Lot, Expiry 6) Eluent B: Lot, Expiry 7) Eluent CV: Lot, Expiry 8) Hemorisisys: Lot, Expiry 9) Calibrator: Lot, Expiry	1) "00000000" (※3) 2) "00000000" 3) -1. "0" 3) -2. "1" 4) -1. "V" 4) -2. "F" 5) "0A1101""2011-01" 6) "0C1112""2011-03" 7) "0D3331""2011-04" 8) "9L1121""2010-12" 9) "CA89C05""2010-05"
	End of record	End of record	<CR> (0x0D)

<Example>

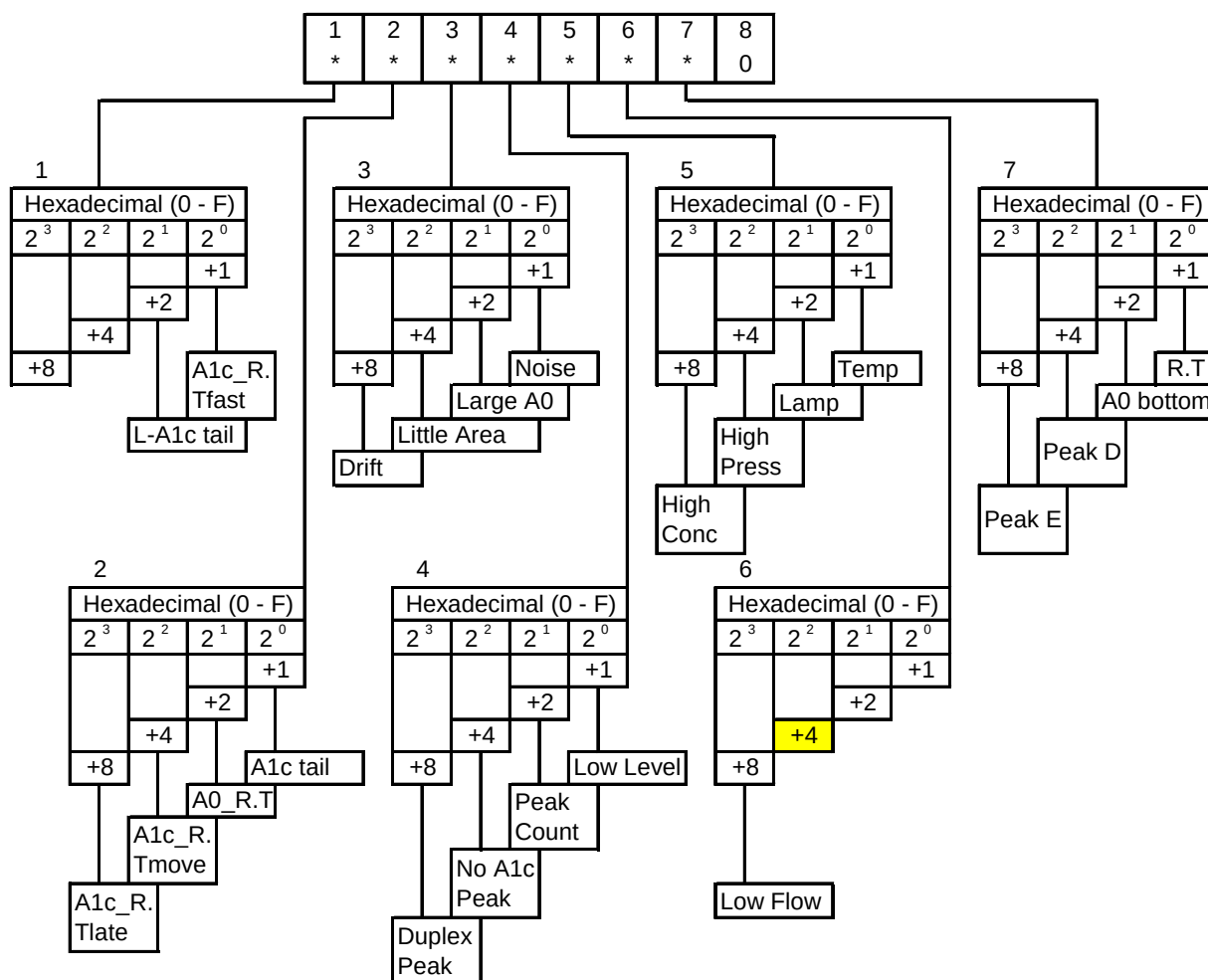
```
O|1|0123456789012345--^0001^01|0001|^^^HbA1c|R|H|00000000^00000000^0^
V^0A1101^2011-0^0C1112^2011-03^0D3331^2011-04^9L1121^2010-12^CA89C05^2010-05<CR>
```

(*1) The maximum digit number for the barcode ID is 18. If it is not met, "-" will be used for the lower position.

(*2) Measurement item consists of four components but the first three are not used.

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[Bit array for error information] (※3)



1 (None)
(None)
L-A1c tail ... L-A1c tail
A1c_R.T fast ... S-A1c retention time (early)

2 A1c_R.T late ... S-A1c retention time (late)
A1c_R.T move . S-A1c R. time fluctuation
A0_R.T ... HbA0 retention time
A1c tail ... S-A1c tail

3 Drift ... Drift baseline
Little Area ... Hb: Low value
Large A0 ... HbA0: Abnormally high value
Noise ... Noise detected

4 Duplex Peak ... Duplex peaks
No A1c Peak .. Cannot detect S-A1c peaks
Peak Count ... Abnormal peak count
Low Level ... Hb: Abnormally low value

5 High Conc ... Hb: Abnormally high value
High Press ... High pressure tube: High pressure
Lamp ... Low optical unit light
Temp ... Temperature control

6 Low Flow ... High pressure tube: Low pressure

7 Peak E ... Abnormal peak (E) detected
Peak D ... Abnormal peak (D) detected
A0 bottom ... HbA0 bottom
R.T ... Retention time

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(3) Result record (Sequence No.1)

One result record is composed of 12 fields.

No.	Field	Description	Remark (Example)
1	Record type	1) Set character	1) "R"
2	Sequence No.	1) Sequence number starting with "1"	1) "1"
3	Measurement item	1) Item name: Set characters (one of below) ^(*1) 1) -1. HbA1c measurement value 1) -2. IFCC measurement value 1) -3. HbF measurement value 1) -4. Total area 1) -5. HbA1ab peak ^(*2) 1) -6. HbA1c peak ^(*2) 1) -7. HbF peak ^(*2) 1) -8. L-A1c peak ^(*2) 1) -9. #A0 peak ^(*2) 1) -10. A0 peak ^(*2) 1) -11. HbS peak ^(*2) 1) -12. HbC peak ^(*2) 1) -13. Peak (no name) ^(*2) 1) -14. Bottom time 1) -15. Baseline contact position (start/end) time 1) -16. Bottom fractionation time HbA1c peak 1) -17. Bottom fractionation time HbF peak 1) -18. Bottom fractionation time HbS peak 1) -19. Bottom fractionation time HbC peak 1) -20. Time for chromatogram scale adjustment ^(*7) 1) -21. Value for chromatogram scale adjustment ^(*7) 1) -22. A/D count value 1) -23. A/D count value 1) -51. A/D count value	1) -1. "^^^ValueHbA1c" 1) -2. "^^^ValueIFCC" 1) -3. "^^^ValueHbF" 1) -4. "^^^AreaTotal" 1) -5. "^^^PeakHbA1ab" ^(*2) 1) -6. "^^^PeakSA1c" ^(*2) 1) -7. "^^^PeakHbF" ^(*2) 1) -8. "^^^PeakLA1c" ^(*2) 1) -9. "^^^Peak#A0" ^(*2) 1) -10. "^^^PeakA0" ^(*2) 1) -11. "^^^PeakHbS" ^(*2) 1) -12. "^^^PeakHbC" ^(*2) 1) -13. "^^^Peak" ^(*2) 1) -14. "^^^TimeBottom" 1) -15. "^^^TimeBase" 1) -16. "^^^TimeSA1c" 1) -17. "^^^TimeHbF" 1) -18. "^^^TimeHbS" 1) -19. "^^^TimeHbC" 1) -20. "^^^TimeGain" ^(*7) 1) -21. "^^^ValueGain" ^(*7) 1) -22. "^^^ADCount1" 1) -23. "^^^ADCount2" 1) -51. "^^^ADCount0"
4	Measurement value	1) Data: digits/symbols (For Item name) 1) -1. HbA1c measurement value (Area ratio) 1) -2. IFCC measurement value (Concentration) 1) -3. HbF measurement value (Area ratio) 1) -4. Total count value 1) -5. HbA1ab peak appearance time, area value, area ratio 1) -6. HbA1c peak appearance time, area value, area ratio 1) -7. HbF peak appearance time, area value, area ratio 1) -8. L-A1c peak appearance time, area value, area ratio 1) -9. #A0 peak appearance time, area value, area ratio 1) -10. A0 peak appearance time, area value, area ratio 1) -11. HbS peak appearance time, area value, area ratio 1) -12. HbC peak appearance time, area value, area ratio	1) -1. "10.0" 1) -2. "101" 1) -3. "10.2" 1) -4. "45000" 1) -5. "121^123451^12.1" 1) -6. "122^123452^12.2" 1) -7. "123^123453^12.3" 1) -8. "124^123454^12.4" 1) -9. "125^123455^12.5" 1) -10. "126^123456^12.6" 1) -11. "126^123456^12.6" 1) -12. "126^123456^12.6" 1) -13. "127^123457^12.7"

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		1) -13. Peak appearance time, area value, area ratio 1) -14. Bottom time ^(*3) 1) -15. Baseline contact position (start/end) time ^(*4) 1) -16. Bottom fractionation time HbA1c peak 1) -17. Bottom fractionation time HbF peak 1) -18. Bottom fractionation time HbS peak 1) -19. Bottom fractionation time HbC peak 1) -20. Time for chromatogram scale adjustment 1) -21. Value for chromatogram scale adjustment 1) -22. A/D count value ^(*5) 1) -23. A/D count value ^(*5) 1) -51. A/D count value ^(*5)	1) -14. "123^234^345" ^(*3) 1) -15. "123^234^345" ^(*4) 1) -16. "123^234" 1) -17. "123^234" 1) -18. "123^234" 1) -19. "123^234" 1) -20. "123" 1) -21. "123^234" 1) -22. "1234^2345^ . ." ^(*5) 1) -23. "1234^2345^ . ." ^(*5) 1) -51. "1234^2345^ . ." ^(*5)
5	Unit	1) Unit: Symbol (For Item name) 1) -1. Area ratio (%) 1) -2. Concentration (mmol/mol) 1) -3. Area ratio (%) 1) -4. Total count value 1) -5. Peak appearance time (second), area value, area ratio (%) 1) -6. Peak appearance time (second), area value, area ratio (%) 1) -7. Peak appearance time (second), area value, area ratio (%) 1) -8. Peak appearance time (second), area value, area ratio (%) 1) -9. Peak appearance time (second), area value, area ratio (%) 1) -10. Peak appearance time (second), area value, area ratio (%) 1) -11. Peak appearance time (second), area value, area ratio (%) 1) -12. Peak appearance time (second), area value, area ratio (%) 1) -13. Peak appearance time (second), area value, area ratio (%) 1) -14. Bottom time (second) 1) -15. Baseline contact position (start/end) time (second) 1) -16. Bottom fractionation time HbA1c peak (second) 1) -17. Bottom fractionation time HbF peak (second) 1) -18. Bottom fractionation time HbS peak (second) 1) -19. Bottom fractionation time HbC peak (second) 1) -20. Time for chromatogram scale adjustment (second) 1) -21. Value for chromatogram scale adjustment (mOD) 1) -22. A/D count value 1) -23. A/D count value 1) -51. A/D count value	1) -1. "%" 1) -2. "mmol/mol" 1) -3. "%" 1) -4. "count" 1) -5. "sec^count^%" 1) -6. "sec^count^%" 1) -7. "sec^count^%" 1) -8. "sec^count^%" 1) -9. "sec^count^%" 1) -10. "sec^count^%" 1) -11. "sec^count^%" 1) -12. "sec^count^%" 1) -13. "sec^count^%" 1) -14. "sec" 1) -15. "sec" 1) -16. "sec" 1) -17. "sec" 1) -18. "sec" 1) -19. "sec" 1) -20. "sec" 1) -21. "mOD" 1) -22. "count" 1) -23. "count" 1) -51. "count"

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7	Abnormal flag	1) Abnormal flag 1) -1. Normal 1) -2. Abnormal	1) -1. NULL (No output) 1) -2. "ERR." (Set characters)
9	Test result status	1) Set character	1) "F"
12	Measurement start date / time	1) Measurement start date / time (YYYYMMDDHHMM): 1) -1 Year, Month, Date, Hour, Minute: 12-digit number After 1) -2. : No output	1) -1. "200607191010" After 1) -2. : NULL (No output)
	End of record	End of record	<CR> (0x0D)

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<Example>

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R|1|^^^ValueHbA1c|3.5|%||||F|||200812151200<CR>
R|2|^^^ValueIFCC|20|mmol/mol||||F<CR>
R|3|^^^ValueHbF|0.5|%||||F<CR>
R|4|^^^AreaTotal|38897|count||||F<CR>
R|5|^^^PeakHbA1ab|4^895^2.3|sec^count^%||||F<CR>
R|6|^^^PeakHbA1ab|6^934^2.4|sec^count^%||||F<CR>
R|7|^^^PeakHbF|8^194^0.5|sec^count^%||||F<CR>
R|8|^^^PeakLA1c|13^467^1.2|sec^count^%||||F<CR>
R|9|^^^PeakSA1c|19^1361^3.5|sec^count^%||||F<CR>
R|10|^^^PeakA0|36^35046^90.1|sec^count^%||||F<CR>
R|11|^^^PeakHbS|47^4844^16.5|sec^count^%||||F<CR>
R|12|^^^PeakHbC|66^4513^15.1|sec^count^%||||F<CR>
R|13|^^^TimeBottom|1^5^8^10^16^31^48|sec||||F<CR>
R|14|^^^TimeBase|0^29|sec||||F<CR>
R|15|^^^TimeSA1c|16^29|sec||||F<CR>
R|16|^^^TimeHbF|9^11|sec||||F<CR>
R|17|^^^TimeHbS|46^48|sec||||F<CR>
R|18|^^^TimeHbC|65^67|sec||||F<CR>
R|19|^^^TimeGain|32|sec||||F<CR>
R|20|^^^ValueGain|10^350|mOD||||F<CR>
R|21|^^^ADCount1|006F^006E^006E^006F^006E^006E^006E^006F^006D^006D^006D^0070^
0072^006E^006D^006E^006E^006D^006E^006E^006E^006D^006D^006D^006D^
006D^006E^006E^006E|count||||F<CR>
R|22|^^^ADCount2|006F^0070^0071^0075^007B^0082^0086^0088^008A^008B^008F^0095^00A2^
00B8^00D4^00F3^010F^0123^0132^013B^0141^0142^0140^013B^0135^012C^
0124^011A^0112^010A|count||||F<CR>
...
R|50|^^^ADCount0|0080^007F^007E^007D^007E^007D^007B^007B^007B^007A^0078^0079^007B^
007E^007B^0078^0078^0079^0079^0079^0079^0078^0078^0078^0077^0077^
0077^0077^0077^0077|count||||F<CR>

```

(*1) Measurement item consists of four components but the first three are not used.

(*2) The maximum number of peaks is 20.

(*3) Bottom time consists of a maximum of 20 components. (All is bottom data.)

The output number varies according to the detected peaks.

(*4) Baseline contact position time consists of a maximum of 20 components. (All is contact position time data.) The output number varies according to the contact positions between the baseline and chromatogram.

(*5) AD count value consists of a maximum of 30 components and is expressed in 4-digit hexadecimal form.

Variant mode: A/D count value consists of 30 records (900 components).

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Fast mode: A/D count value consists of 16 records (480 components).

(*7) “Time for chromatogram scale adjustment” indicates “switching time” in the variant mode but “HbA1c peak time” in the Fast mode.
In the Variant mode, two values are shown for chromatogram scale adjustment. Whereas in the Fast mode, there is only one value for chromatogram scale adjustment. As such, two identical values (“19^19”) are output in the Fast mode.